DEIS Lab 2: Basic Robotics 2
DT8007 VT26, Halmstad University

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Introduction

In this lab we programmed an Arduino UNO R4 WiFi, did some basic hardware prototyping, flew a Ryze Tello drone (manually and from Python), and got a SparkFun RedBot ground robot to move.

1 Arduino

1.1 Getting started

We installed the UNO R4 board package in the Arduino IDE, selected the board and port, and uploaded the Blink example.

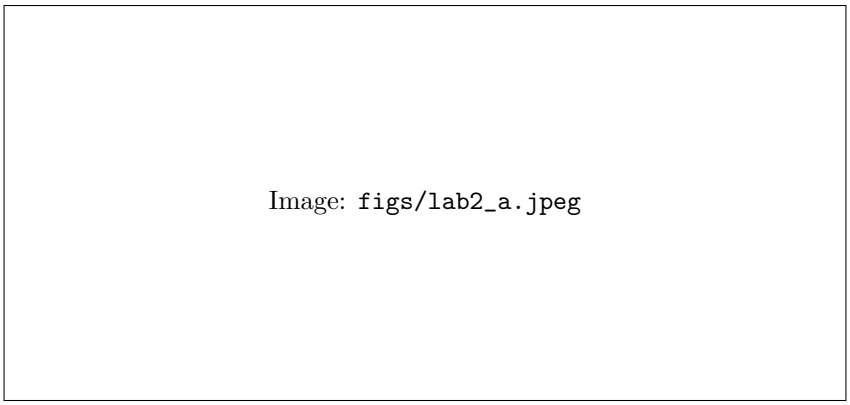


Figure 1: Blink example running on the Arduino UNO R4 WiFi.

1.2 Connecting to WiFi

We uploaded the `SimpleWebServerWiFi` example with the course router's SSID and password, opened the IP address from the Serial Monitor in a browser, and turned the LED on and off (`/H` and `/L`).

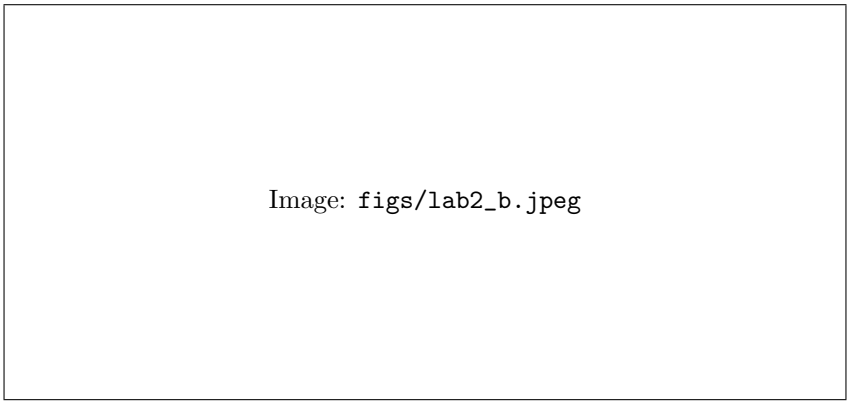


Figure 2: LED turned on from the browser over WiFi.

2 Prototyping

Together with the supervisor we did simple tasks with a multimeter, soldering iron and drill.

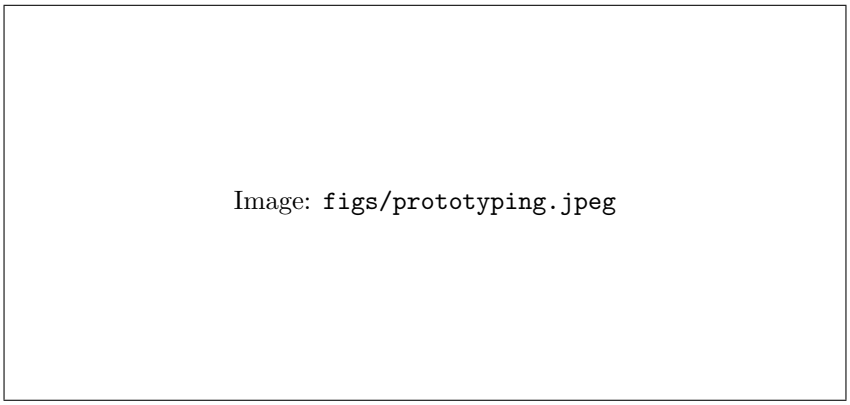


Figure 3: Prototyping souvenir photo.

3 Drone

3.1 Manual flying

Using the Ryze Tello app we took off, rotated left and right, flew up and down, forward and backward, strafed left and right, and landed.

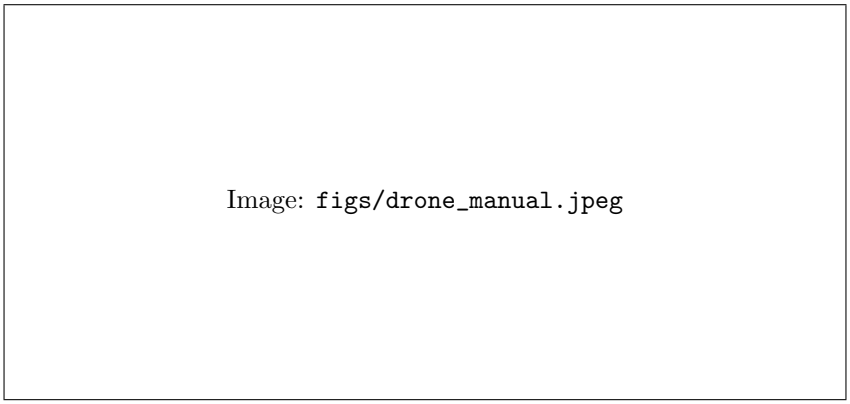


Figure 4: Manual flying souvenir photo.

3.2 Control via Python

We used Shiva's script[1] to get the video feed and control the drone from a laptop, and saved an image from the drone's camera.



Figure 5: Image captured from the drone's camera via Python.

4 Ground robot

We assembled the RedBot (main board, IR line sensors, wheel encoders, alkaline battery pack) and worked through the SparkFun tutorials 1–3 (basic motions), 6 (line following) and 7 (encoders)[2].

[1] https://github.com/SHiTU23/Tello_drone/blob/main/src/control_and_capture.py
[2] https://learn.sparkfun.com/tutorials/experiment-guide-for-redbot-with-shadow-chassis

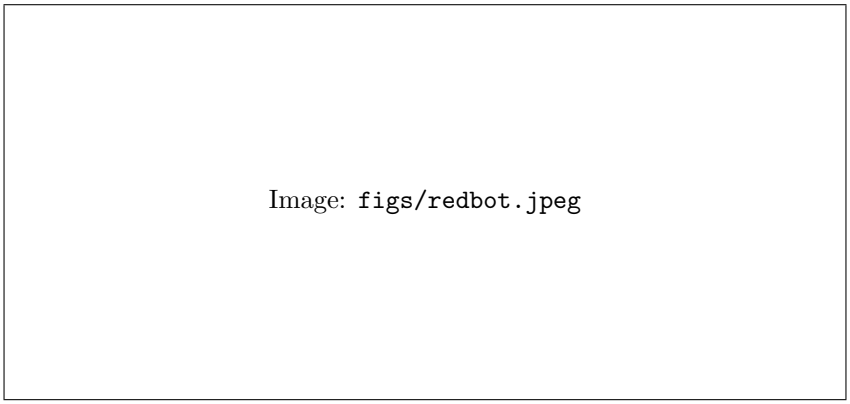


Figure 6: Assembled RedBot.

Comment. Since we did not have tape to make a line-following track, we used the Arduino UNO R4 WiFi as the main controller instead. It was connected to the RedBot through RX/TX (UNO TX $\rightarrow$ RedBot RX, UNO RX $\leftarrow$ RedBot TX, common GND, 9600 baud) and powered by a mobile power bank. The UNO serves a web page with W/A/S/D controls and forwards each command to the RedBot. The RedBot drives the motors and sends back the IR line-sensor and encoder readings, which are shown live on the page. This is how we tested the motions, the line sensors and the encoders. As a safety measure, the RedBot stops if no command arrives for 500 ms.

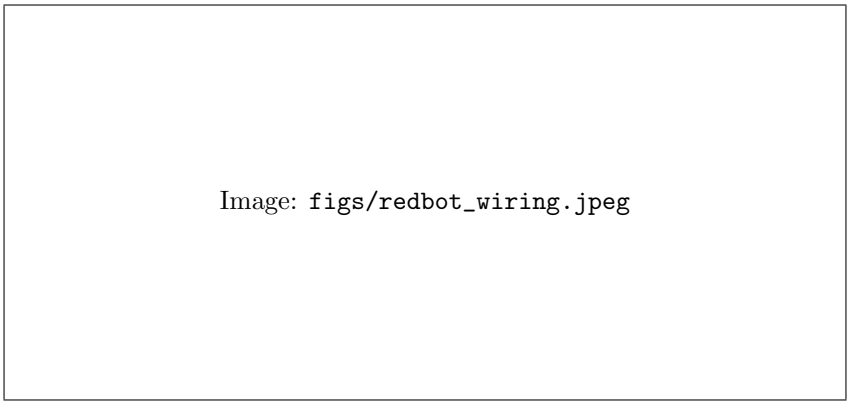


Figure 7: UNO R4 WiFi (on power bank) connected to the RedBot over RX/TX.

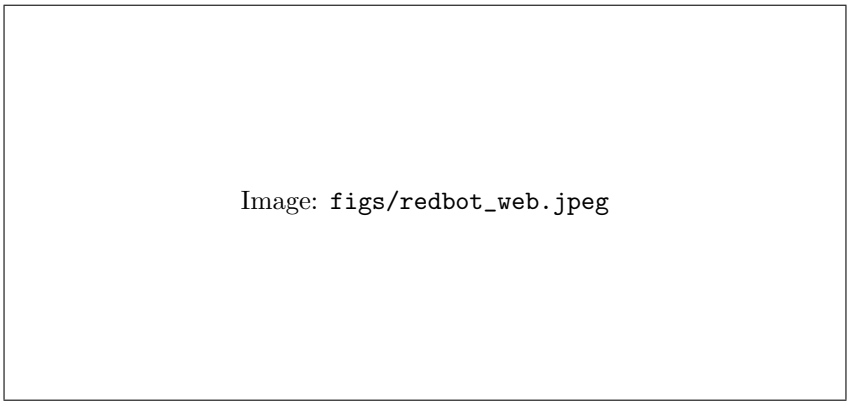


Figure 8: Web control page with live IR and encoder readings.

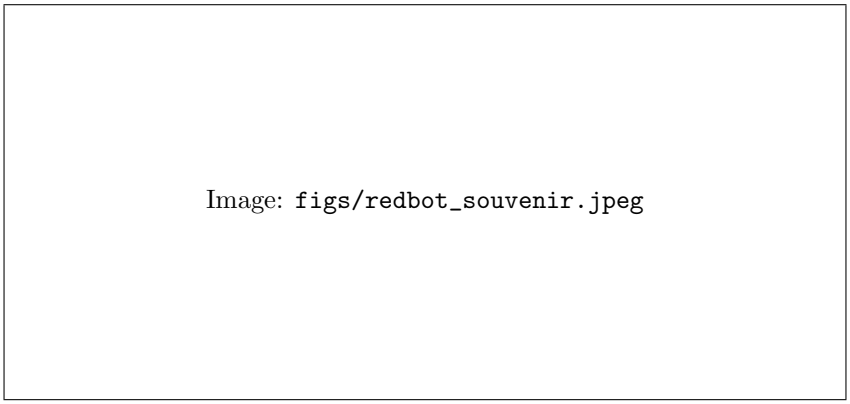


Figure 9: Ground robot souvenir photo.

Contribution

Activity	Binay Kumar Sah	Kabir Tamari
(a) Arduino	50%	50%
(b) Prototyping	50%	50%
(c) Drone	50%	50%
(d) Ground robot	50%	50%
Report	50%	50%

Use of AI

An AI assistant (Claude) was used to help write the code for the ground robot task and to draft this report.